

# Rajeev Jain

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Principal Specialist, Research Software Engineering | AI Systems and Verification | HPC | Scientific Computing

## Profile

Principal research software engineer, at Argonne since 2009, turning research prototypes into production systems across AI infrastructure, scientific computing, and HPC. Lead developer of UXarray, a Python library for unstructured climate-grid analysis used at DOE labs and NCAR, including an MCP server that gives AI agents typed, scoped, provenance-tracked access to Earth-system meshes on HPC clusters via Globus Compute — the alternative to handing a team general-purpose chat and calling it a strategy. Equally focused on verifying what such systems produce: caught silent numerical drift, hidden heap allocation, and unreachable vendor throughput figures in code and benchmarks that read as correct. Ran 10,000+ cancer-AI training experiments on Summit; ported the Pangu-Weather climate emulator to Intel XPUs on Aurora. Two R&D 100 Awards (CANDLE 2023, FLASH-X 2022).

## Professional Experience

### Argonne National Laboratory

2009 – Present

*Principal Specialist, Research Software Engineering*

*Mathematics and Computer Science Division, Chicago, IL*

- **UXarray and MCP server.** Lead developer of a Python library for unstructured climate grid analysis used at DOE labs and NCAR. Designed grid I/O for ESMF / MPAS / SCRIP / HEALPix and conservative zonal averaging. Built the UXarray MCP server: 20+ typed tools letting AI agents inspect, subset, visualize, and run analysis on Earth-system meshes locally or on HPC clusters (Argonne Improv, UCAR Derecho) via Globus Compute, with structured provenance. Presented at SciFM26.
- **AI code and benchmark verification.** Independent measurement applied to AI tooling and machine-assisted code. Published a reproducible harness for local LLM inference: bytes-per-token parsed from GGUF tensor tables rather than file size (which overstates 64% on gathered-embedding models), and memory bandwidth measured at 340 GB/s against an unreachable 400 GB/s datasheet pin rate — showing four different architectures all land within 36–41% of bandwidth, so throughput is set by model size and runtime, not architecture. In numerical Python, traced 0.68 m of drift on a production Earth mesh to catastrophic cancellation inside a spherical cross product — a difference of nearly equal products, not a running sum — and proposed error-free-transformation arithmetic for the kernel (merged upstream, UXarray PR #1513); recovered a  $1.67\times$  speedup by removing heap allocation a faithful C++-to-Python port had silently introduced.
- **Pangu-Weather on Aurora.** Ported the Pangu-Weather climate emulator to Aurora's Intel XPU stack (60,000+ GPUs): PMIX/PALS launcher mapping, XPU/CUDA device branching, device-aware AMP (GradScaler on CUDA, bf16 on XPU), first stable DDP baseline. Established smoke-test and queue-aware bring-up scripts separating launcher, backend, and device-placement failures. Measured one-node baseline: 12 XPU ranks under DDP,  $\sim 12$  s steady-state epochs after a warmup epoch  $8\times$  longer.
- **CANDLE / IMPROVE.** Built CANDLE/Supervisor HPO workflow (Swift/T, mlrMBO, DEAP, Hyperopt/TPE); ran 10,000+ cancer drug response experiments on Summit, Theta, and Cori. Applied noise injection and counterfactual analysis to NT3 tumor classification; identified PLOD2, RGS5, and TP53I13 as decision-boundary drivers. Led IMPROVE: `improvelib` Python package with GitHub Actions CI/CD (unit tests, Docker, Parsl CSA) and UNO dual-branch neural network for cross-study benchmarking. 15+ researchers across Argonne, LLNL, ORNL. R&D 100 Award, 2023.
- **FLASH-X.** I/O and compression lead for a million-line multiphysics engine. Async HDF5 with Argobots plus SZ3 / ZFP compression reduced checkpoint overhead 40–70% on Summit with 50%+ storage savings. Enabled cross-checkpoint restart between AMReX and Paramesh. R&D 100 Award, 2022.
- **MeshKit and Urban ECP.** PI for MeshKit (DOE NEAMS, 2009–2016): C++ toolkit for reactor mesh generation; parallel CoreGen produced a 101M-element MONJU mesh on 712 processors in about 7 minutes. Led Urban ECP: coupled WRF/HRRR  $\rightarrow$  EnergyPlus  $\rightarrow$  Nek5000 workflow with full 3D downtown Chicago domain. Published in *Journal of Building Performance Simulation*, 2020.

### University of Chicago

2023 – Present

*Staff At-Large*

*Consortium for Advanced Science and Engineering (CASE), Chicago, IL*

- Joint appointment supporting cancer pharmacogenomics and Earth science software collaborations; mentoring

graduate students on software design and HPC workflows.

## Arizona State University

Research and Teaching Assistant

2007 – 2009

Structural and Computational Mechanics Lab, Tempe, AZ

- FEM research on blast mitigation and design optimization; supported structural engineering coursework.

## Earlier Experience

Project Engineer, Wipro Technologies / Engineering Intern, Tata Motors

2005 – 2007

India

## Technical Skills

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**Languages:** Python, C++, Fortran, Bash, R, SQL

**ML / AI:** PyTorch, TensorFlow, NumPy, Pandas, Xarray, Scikit-learn; Bayesian HPO (mlrMBO, Hyperopt/TPE), genetic algorithms (DEAP)

**AI Systems:** MCP server design (typed tools, scoped access, structured provenance), Claude API, LLM tool-use and agent workflows, local inference (llama.cpp, GGUF, quantization), Globus Compute for HPC-backed agents

**Correctness / Perf:** Numerical analysis (compensated summation, error-free transformations), roofline and bandwidth analysis, profiling, review of ported and machine-generated numerical code, reproducible benchmark design

**HPC / Systems:** MPI, OpenMP, HDF5, NetCDF, MOAB, Argobots, SZ3/ZFP, Intel XPU / CUDA, PBS/SLURM

**Workflow / DevOps:** GitHub Actions, Docker, Singularity, Parsl, Swift/T, release automation, CI/CD, Globus Compute

**Domains:** Climate and weather modeling, cancer pharmacogenomics, computational physics, reactor mesh generation

## Selected Publications

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- **Jain, R.** and Jacob, R. “[Beyond Tool Execution: Evaluating Scientific MCP Interfaces with UXarray.](#)” *1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026*, Naples — accepted, to be presented. Artifacts: [doi:10.5281/zenodo.21463232](https://doi.org/10.5281/zenodo.21463232).
- Partin, A., ..., **Jain, R.**, et al. “[Benchmarking community drug response prediction models.](#)” *Briefings in Bioinformatics*, 2026.
- **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. “[Enabling Data Reduction for Flash-X Simulations.](#)” *SC24 / DRBSD-10*, 2024.
- **Jain, R.**, Wozniak, J. M., et al. “[Cross-HPO: Optimizing Neural Networks for Cancer Drug Response.](#)” *CAFCW at SC24*, 2024.
- Wozniak, J. M., **Jain, R.**, et al. “[CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research.](#)” *BMC Bioinformatics*, 2018.

Full list: [Google Scholar](#).

## Awards, Recognition, and Service

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- Program committee member, 1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026.
- R&D 100 Award, CANDLE (2023) | R&D 100 Award, FLASH-X (2022).
- DOE SBIR Phase I & II (RGG reactor mesh generator; Kitware, 2014–2017).
- ATPESC Scholar, Argonne Training Program on Extreme-Scale Computing (2015).
- Best Paper Award, International Meshing Roundtable (2010).

## Education

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### University of Chicago

M.S. in Computer Science

2020

### Arizona State University

M.S. in Structural Engineering

2009

Graduate Fellowship Recipient

### Indian Institute of Technology (ISM) Dhanbad

B.Tech. in Mechanical Engineering

2006